
```

n = 10;

A = rand(n, n);
for i = 1:n
    % ne asiguram ca elementul de pe diagonala este mai mare decat
    % suma modulelor celorlalte elemente de pe linie.
    suma_linie = sum(abs(A(i, :))) - abs(A(i, i));
    A(i, i) = suma_linie + rand() + 1;
end

x_exact = (1:n)';

b = A * x_exact;

tol = 1e-6;
max_iter = 1000;

[x_jacobi, iter_jacobi] = metoda_jacobi(A, b, tol, max_iter);
eroare_jacobi = norm(x_jacobi - x_exact, inf);
fprintf('Jacobi:          Iteratii = %3d, Eroare = %e\n', iter_jacobi,
eroare_jacobi);

[x_gs, iter_gs, ~] = metoda_sor(A, b, 1, tol, max_iter);
eroare_gs = norm(x_gs - x_exact, inf);
fprintf('Gauss-Seidel: Iteratii = %3d, Eroare = %e\n', iter_gs, eroare_gs);

[x_sor, iter_sor, w_o] = metoda_sor(A, b, [], tol, max_iter);
eroare_sor = norm(x_sor - x_exact, inf);
fprintf('SOR (w=%.3f): Iteratii = %3d, Eroare = %e\n', w_o, iter_sor,
eroare_sor);

Jacobi:          Iteratii =  54, Eroare = 3.452521e-07
Gauss-Seidel: Iteratii =  12, Eroare = 6.223258e-08
SOR (w=1.191): Iteratii =  18, Eroare = 2.834781e-07

```

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